# Island Counter

Category: Graph Algorithms | Language: Python 3

## 1. Aim

Count the number of separate islands in a rectangular map. In the map, 1 means land and 0 means water. Land cells connect only vertically and horizontally, not diagonally.

## 2. BFS method

The program scans each cell. When it finds a land cell that has not been visited, it increases the island count and starts a Breadth-First Search (BFS) using a queue.

The search visits all reachable land cells in four directions: up, down, left, and right. These cells belong to the same island, so they are marked as visited and are not counted again.

## 3. Example

| Grid row | Explanation |
|---|---|
| 11000 | Top-left island |
| 11000 | Same island |
| 00100 | Middle island |
| 00011 | Bottom-right island |

**Program output:** Number of islands: **3**.

## 4. Complexity

If there are R rows and C columns, each cell is checked a constant number of times. Therefore time complexity is **O(R × C)**. In the worst case, the visited set and BFS queue can take **O(R × C)** space.

## 5. Testing and conclusion

I tested a grid with three islands, a map with no land, a map containing only land, and a diagonal-only case where the two land cells should count as two islands. All unit tests passed.

Conclusion: BFS makes it easy to find connected areas in a grid. This is one way graph algorithms can solve map-like problems.